

Multilingual Email Assessment - LLM Fine-Tuning Challenge

Problem Statement

You are provided with a **multilingual email dataset consist of email of 5 different languages**.

Each email sample is associated with:

- **Grammar Rating** — a continuous score representing the grammatical quality of the email.
- **Content Rating** — a continuous score representing the quality/relevance of the email content.
- **CEFR Level** — a 6-class ordinal English proficiency level.

Your task is to develop an **LLM-based modeling solution** that can predict these assessment outcomes from the email text.

The objective is not to prescribe a particular architecture or training methodology. You are encouraged to explore and implement the approach you believe is most appropriate.

Your Task

Finetune an LLM or any other AI model that takes an email as input and predicts:

1. **Grammar Score**
2. **Content Score**
3. **CEFR Level**

You may use **one model or multiple models**, and CEFR prediction can be:

- modeled as an independent prediction task, or
- derived/predicted using the Grammar and Content predictions as additional signals.

You have complete freedom to decide how these tasks should be modeled. You may also choose the LLM/any other model architecture, tokenizer, training strategy, loss functions, hyperparameters, data preprocessing and inference strategy. The emphasis is on **your reasoning and ability to develop an effective solution**, rather than following a prescribed implementation.

Training Data

You need to create your own **training data** for this task. You are free to augment or extend the training data using **publicly available open-source datasets, synthetic data, data**

augmentation techniques, or other legally accessible data sources that are relevant to the task.

The provided **test set** will be used as the final evaluation set. Candidate **must not train or fine-tune their model on the test-set labels**.

You may also generate synthetic examples using LLMs or other methods, provided that the methodology is clearly documented. You are responsible for designing the data preparation, augmentation, sampling, and labeling strategy that you believe will best improve model performance.

Important: The provided test set must be treated as strictly held out. Test-set samples or labels must not be used for training, fine-tuning, synthetic data generation, prompt construction, or any other process that could result in data leakage.

CEFR Prediction

CEFR is a **6-class ordinal variable**.

The six levels should be treated as ordered proficiency levels rather than completely independent classes.

You may therefore formulate CEFR prediction as:

- standard multi-class classification,
- ordinal classification,
- regression followed by discretization,
- or any other approach you believe is appropriate.

You may also use the predicted Grammar and Content scores as inputs/features for CEFR prediction.

Acceptance Criteria

The final solution will be evaluated on the **provided held-out test set**.

Your solution should satisfy **all** of the following criteria:

Task	Metric	Ideal Acceptance Criteria
Grammar	Pearson Correlation	≥ 0.60
Content	Pearson Correlation	≥ 0.60

CEFR	Pearson Correlation	≥ 0.60
CEFR	Exact Match Accuracy	$\geq 30\%$

CEFR Evaluation

For CEFR, both **Pearson correlation** and **exact-match accuracy** will be considered.

Since CEFR is ordinal, the correlation evaluates whether the model captures the ordering and relative proficiency differences, while exact match evaluates whether the model predicts the correct CEFR level.

For example, if the ground-truth CEFR is **B2** and the model predicts **B1**, this is counted as an incorrect exact match even though the prediction is only one level away.

Evaluation Protocol

The provided test set will be used as the final evaluation set.

Candidates should **not train or fine-tune their model on the test-set labels**.

The final evaluation will be performed using the candidate's submitted inference pipeline/model on the held-out test data.

The evaluation will independently calculate:

- Grammar Pearson correlation
- Content Pearson correlation
- CEFR Pearson correlation
- CEFR exact-match percentage

Deliverables

Please provide:

1. **Source code** for training/fine-tuning and inference.
2. Prepared Training Data
3. **Trained model or instructions to reproduce the model.**
4. **README/documentation** describing:
 - Model/LLM selected
 - Data preparation and preprocessing method along with data sources
 - Fine-tuning strategy

- Training configuration
- Loss/objective functions
- Hyperparameters
- Evaluation methodology

5. **Validation results** and analysis of your approach.

6. A short explanation of the key design decisions, conclusions and any additional experiments performed.

What We Will Evaluate

The primary requirement is that the solution meets the specified performance thresholds.

However, we will also consider:

- Quality of the Training data preparation approach
- Quality of the modeling approach
- Understanding of the multilingual setting
- Appropriate use of LLM fine-tuning
- Handling of ordinal CEFR labels
- Experimental methodology
- Data leakage prevention
- Error analysis
- Reproducibility
- Clarity of technical reasoning
- Computational efficiency

There is **no prescribed model architecture or fine-tuning technique**. The candidate is encouraged to experiment and justify the approach that they believe provides the best solution.

Key Objective

Build the best multilingual email assessment model you can, subject to the evaluation criteria above.

You are free to use any reasonable LLM, fine-tuning strategy, architecture, loss function, feature engineering, or combination of models to achieve the required performance.

Score Rubrics

Content

Score	Rubric
0	The email showed little to no effort by the candidate. It left out critical information, closely imitated the prompt without adjustments, and/or was very short or off topic.
1	The email briefly mentioned the task or purpose for communication in a few lines but did not provide any details.
2	The email clearly stated the task or purpose of the communication and provided detailed information. Some critical information, such as important details, was missing.
3	The email clearly stated the task or purpose of the communication and provided almost all of the required information. Only a few minor details were missing from the response.
4	The email was complete. It included a clear statement of the task or purpose of communication, and all information required to address it.

Grammar

Score	Rubric
0	The email was off topic, closely imitated the prompt without adjustments, or was less than 3 sentences.
1	The email included many sentence fragments (sentences which did not include a subject or predicate). The email had several severe spelling or grammar errors, making it difficult to understand the meaning.
2	The email had few sentence fragments but included major errors in spelling or grammar. These issues, at times, changed or muddled the meaning of sentences.

3	The email had sentences which were properly structured but still contained observable spelling or grammatical errors. Improper spelling or grammar made the sentences difficult to understand and/or unpolished.
4	All sentences in the email were properly structured, including only some small issues in spelling and grammar. At some points, grammatical inconsistencies or spelling errors detracted from the email's polish, though they did not lead to ambiguity in meaning.
5	All sentences in the email were properly structured, with few to no issues in grammar. There were no or only negligible errors in the appropriate usage of spelling and relevant grammatical constructs. If (minimal) errors were present, they did not lead to any ambiguity in meaning or detract from the email's overall purpose.

CEFR

Level	Rubric
C2	The candidate can write a clear, smoothly flowing, and fully engaging email that expresses complex and relevant ideas and details precisely. The parts of the task and the purpose and key details of the email were properly discussed with no missing details. Their writing is highly organized with seamless transitions and sophisticated use of cohesive devices. They exhibit a professional structure with clear and logical flow which helps the reader to notice and remember significant points. Their writing uses extensive, nuanced, and precise vocabulary, with effortless and proper use of idiomatic and technical language. They consistently maintain a high degree of grammatical accuracy and use complex

	sentence structures naturally and appropriately. They can communicate complex and nuanced ideas with precision and ease, and can adapt their style and tone effortlessly to suit the context, purpose, and audience.
C1	The candidate can write a clear, well-structured, and details email, which underlines relevant information and expands and supports the information in it at some length with subsidiary points and relevant detail. The parts of the task and the purpose and key details of the email were properly discussed with very little to no missing details. Their writing is highly organized with logical progression and sophisticated use of cohesive devices. They use extensive and varied vocabulary, including technical terms and idiomatic expressions. They can convey subtleties and shades of meaning. Their writing has a high degree of grammatical accuracy. They are able to use a wide range of complex structures correctly and naturally. They can communicate complex ideas precisely and using a language style that is adapted to different contexts, audiences, and purposes.
B2	The candidate can write a clear, well-structured, and detailed email that is well-organized with logical flow. The parts of the task and the purpose and key details of the email were properly discussed but with some minor missing details. They can use a variety of cohesive devices to enhance cohesion. Their writing has a broad range of vocabulary and they show a good command of idiomatic expressions and nuanced language. Their writing shows good control of complex sentence structures, with occasional grammatical errors that do not

	hinder communication. They can express their ideas clearly and effectively, and can adopt the tone and level of formality according to the context and the recipient.
B1	The candidate can write an email that provides some details and explanations. Their writing follows an organized structure with clear sections. The parts of the task and the purpose and key details of the email were adequately discussed but with some missing details. They can use basic cohesive devices to link a series of shorter discrete ideas into a linear sequence. Their writing has a wider range of vocabulary, including some less common words and phrases. They are able to paraphrase and use synonyms. They can use a variety of simple and complex sentence structures. Errors occur in their writing but do not impede understanding, and they can convey messages clearly enough for the reader to understand without much difficulty.
A2	The candidate can write a simple email using a series of simple phrases and sentences linked with simple connectors like "and", "but", and "because". The parts of the task and the purpose and key details of the email were very brief. Their writing has a basic structure with an introduction, body, and closing. They can use basic vocabulary and basic grammar and sentence structures which contain frequent errors. The reader may need to guess the meaning of some parts of the message due to these errors.
A1	The candidate can write a short, simple email using basic isolated phrases and sentences. The parts of the task and the purpose and key details of the email are not adequately discussed. Their writing is poorly structured and lacks coherence and logical flow. The

candidate is only able to use extremely basic vocabulary, using a limited range of words and phrases. The sentences they produce are often incomplete or fragmented and contain frequent grammatical errors. This requires significant effort from the reader to understand the message being communicated.